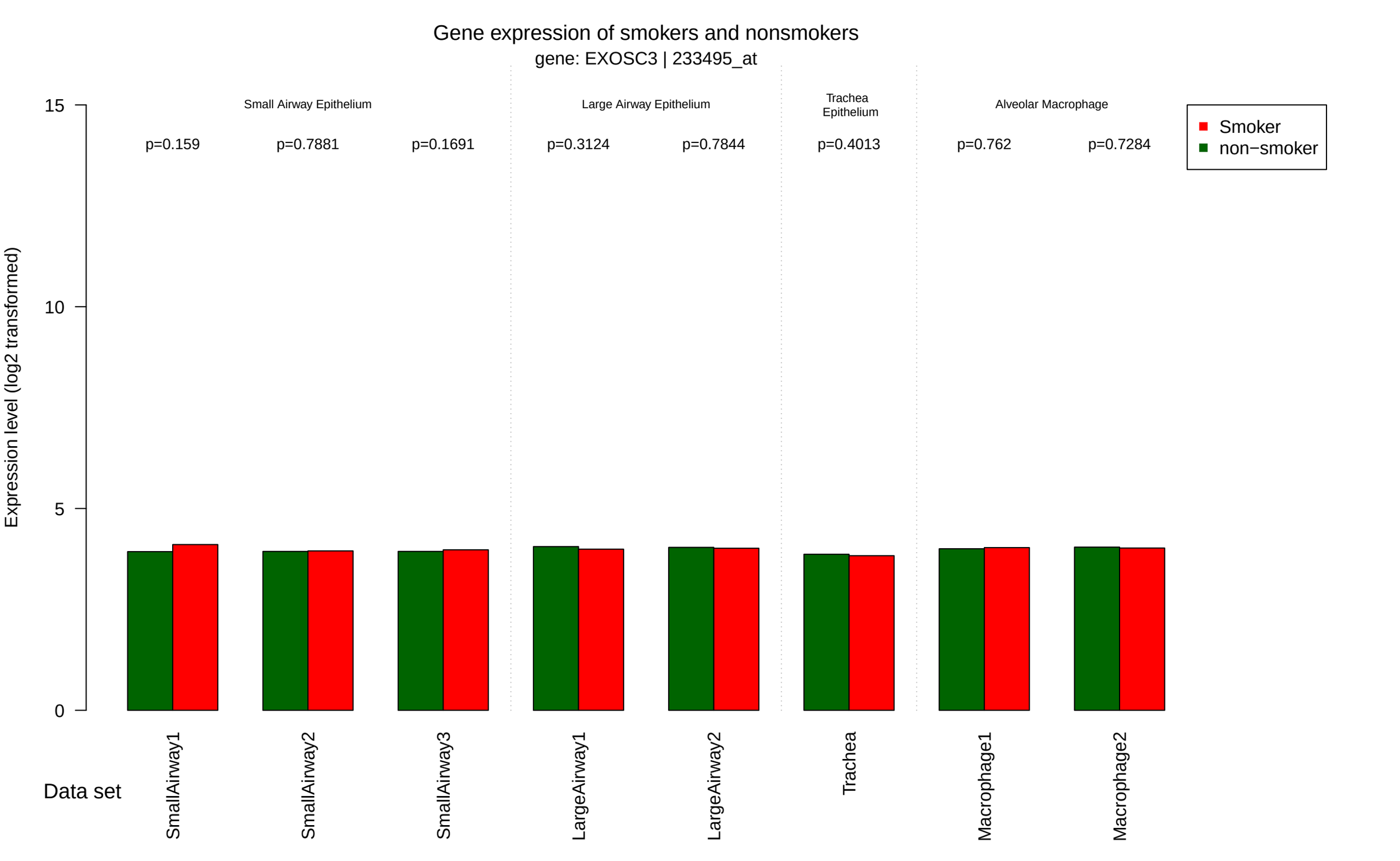
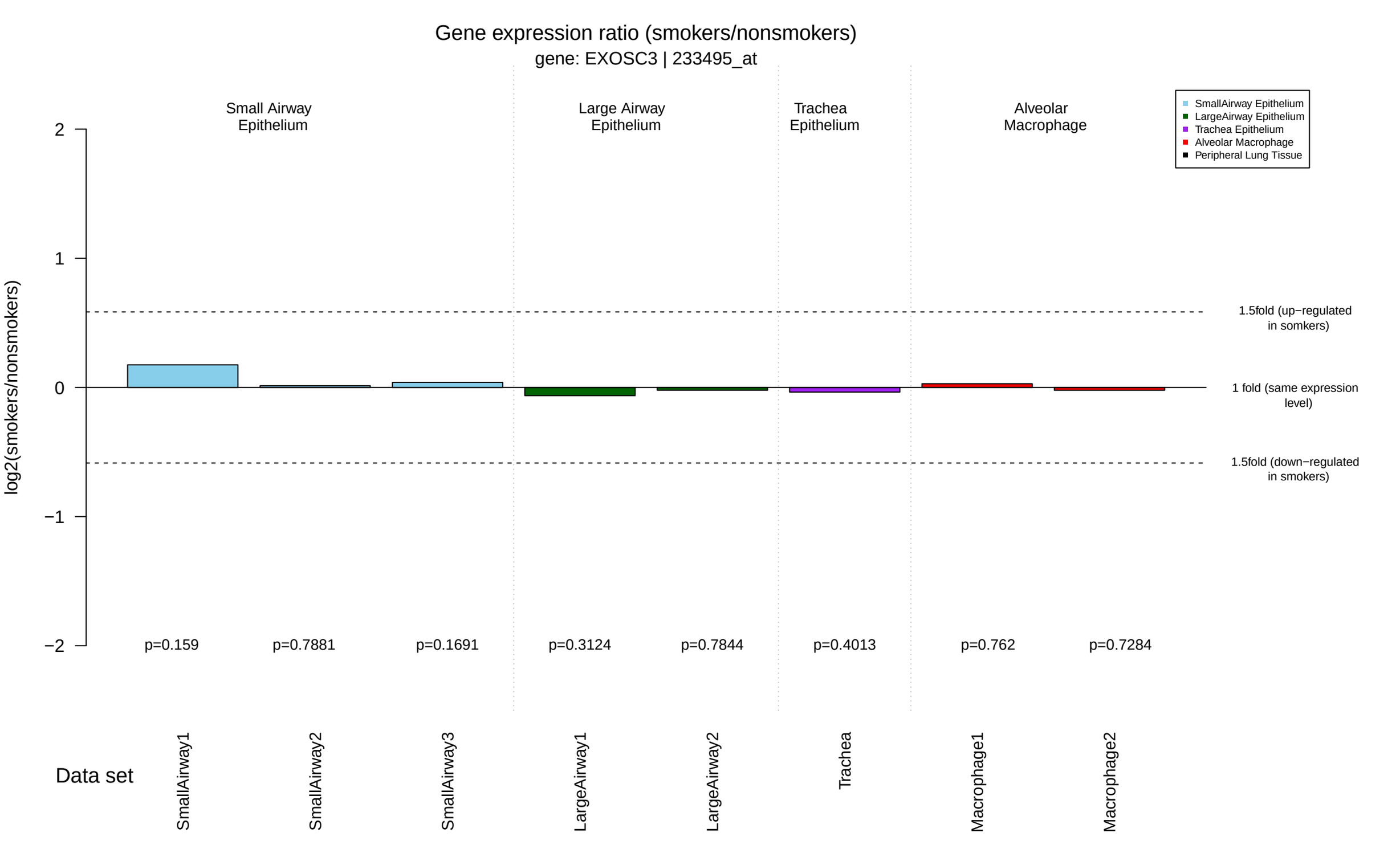
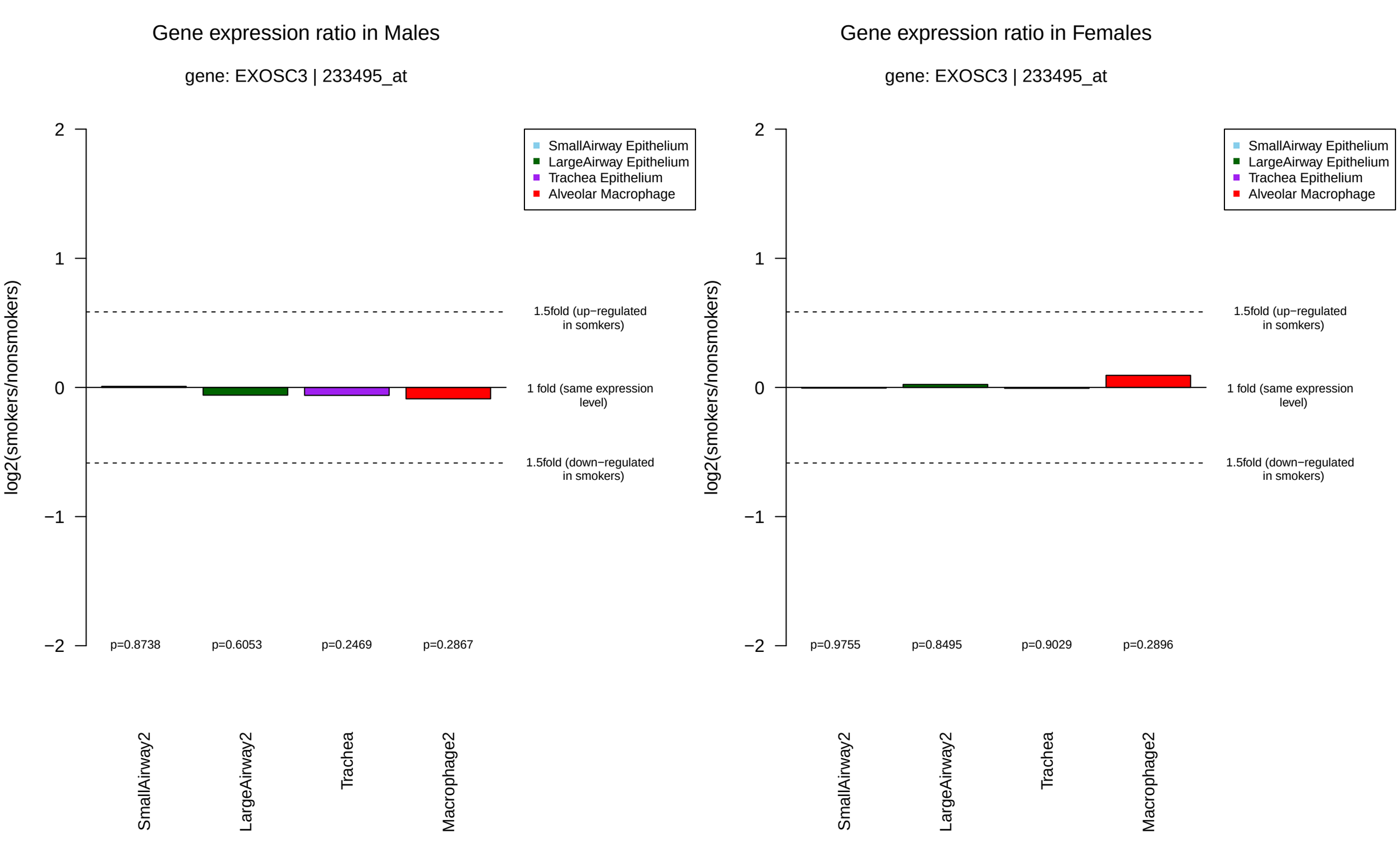
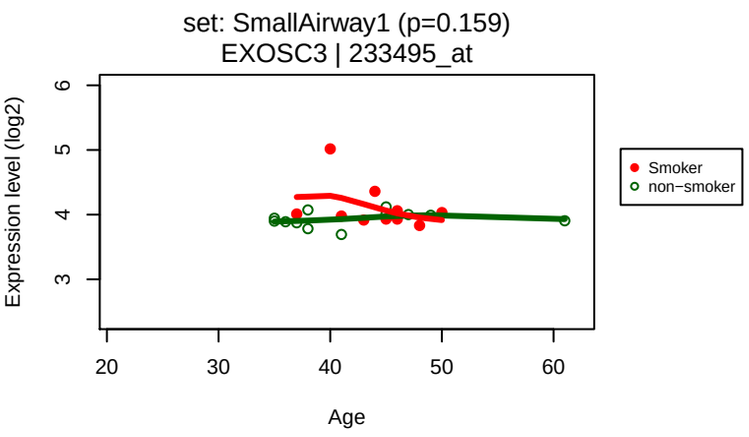
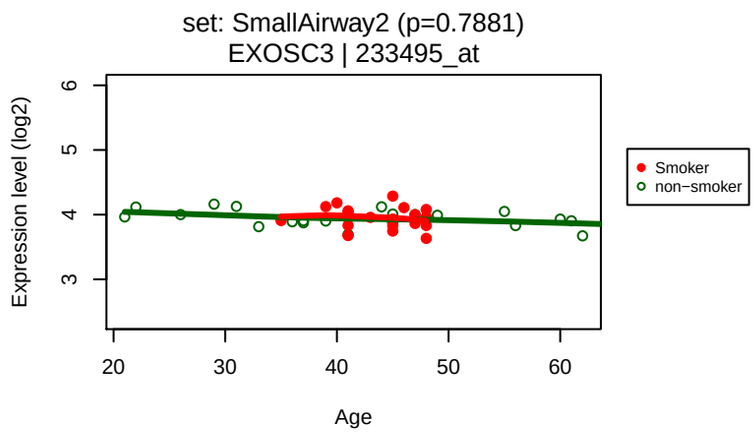
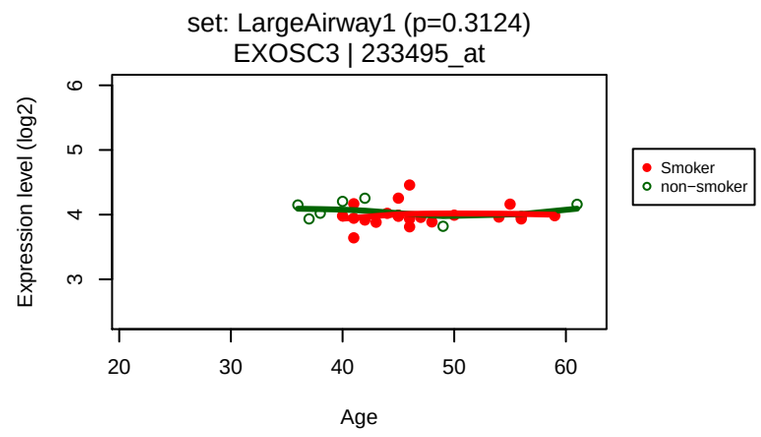

Gene expression of smokers and nonsmokers
gene: EXOSC3 | 233495_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.159
p=0.7881
p=0.1691
p=0.3124
p=0.7844
p=0.4013
p=0.762
p=0.7284
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

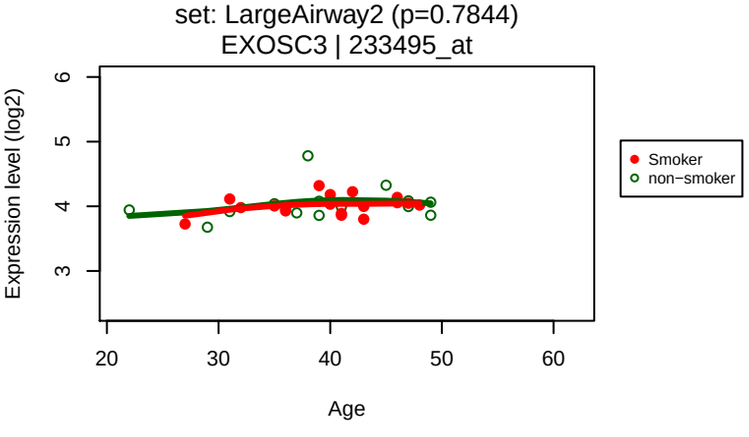

Gene expression ratio (smokers/nonsmokers)
gene: EXOSC3 | 233495_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.159
p=0.7881
p=0.1691
p=0.3124
p=0.7844
p=0.4013
p=0.762
p=0.7284
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

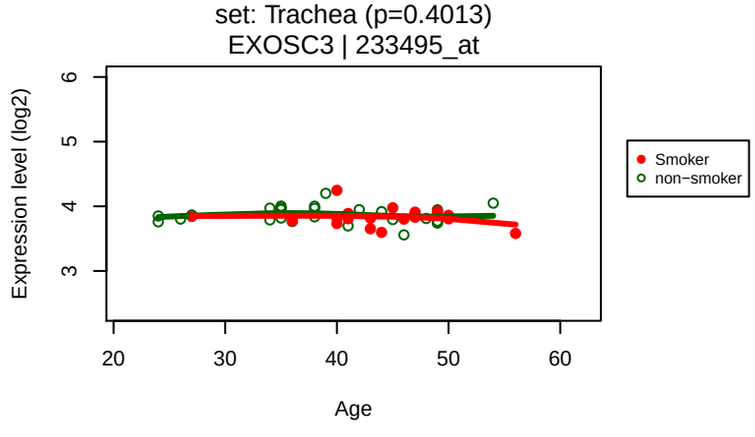

Gene expression ratio in Males
gene: EXOSC3 | 233495_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8738
p=0.6053
p=0.2469
p=0.2867
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EXOSC3 | 233495_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9755
p=0.8495
p=0.9029
p=0.2896
SmallAirway2
LargeAirway2
Trachea
Macrophage2

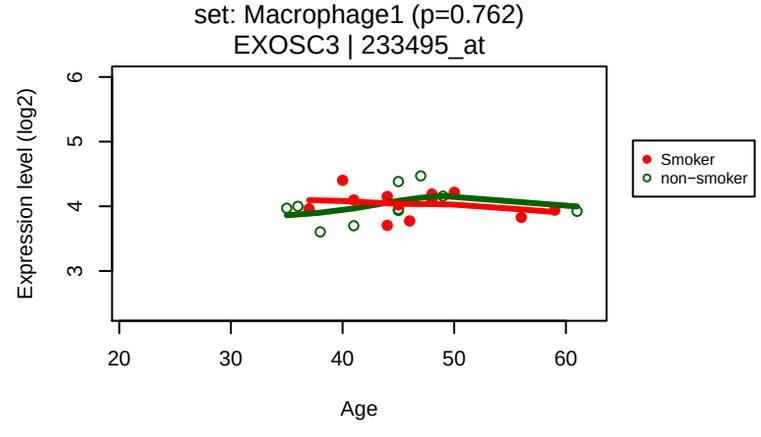

set: SmallAirway1 (p=0.159)
EXOSC3 | 233495_at
Expression level (log2)
Age
Smoker
non−smoker

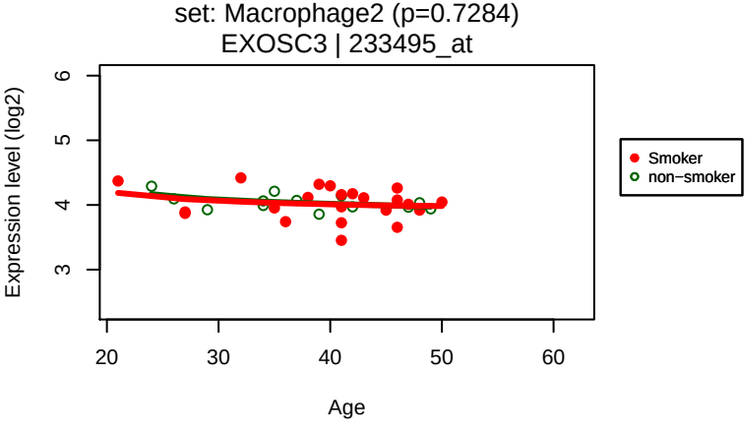

set: SmallAirway2 (p=0.7881)
EXOSC3 | 233495_at
Expression level (log2)
Age
Smoker
non−smoker

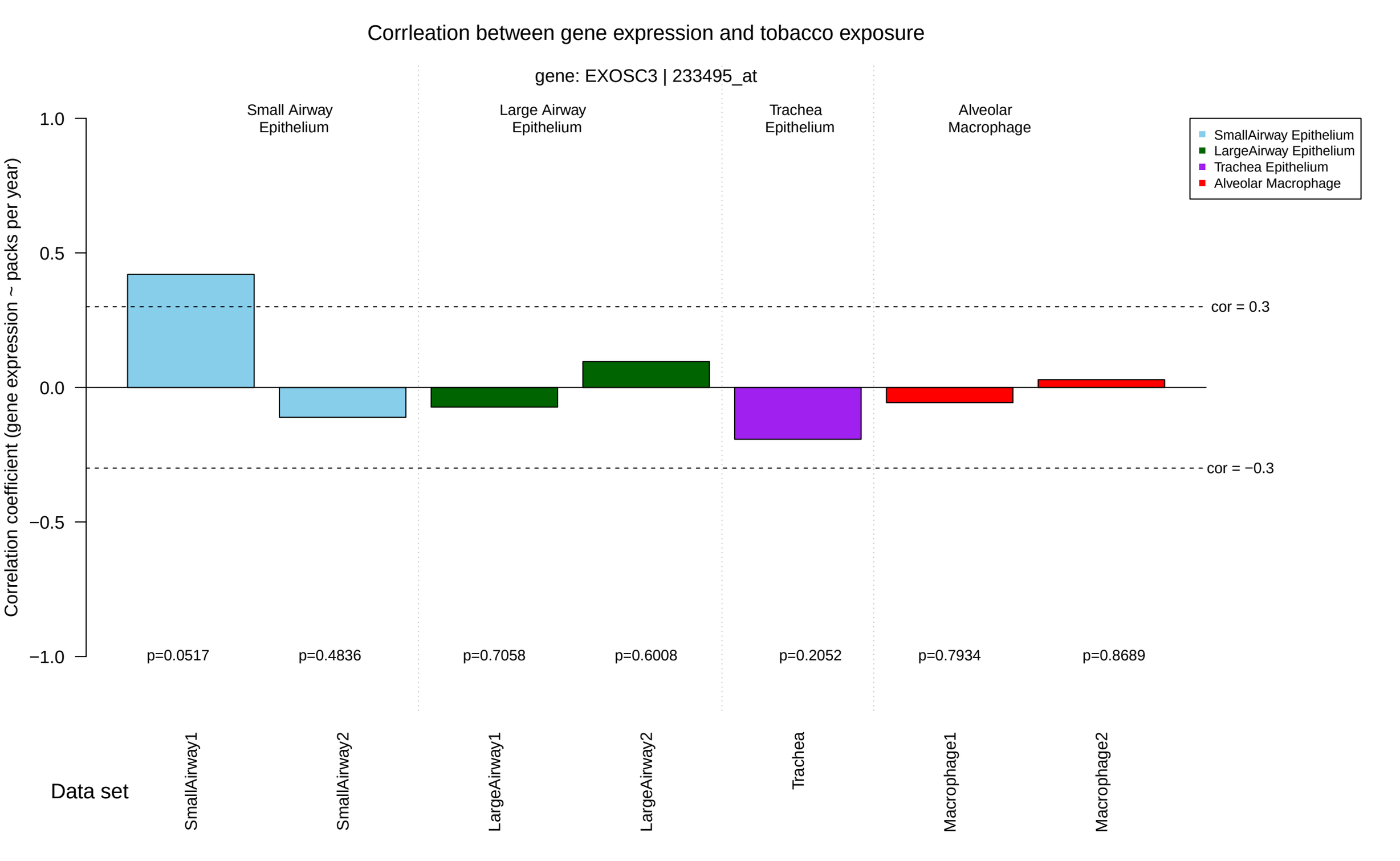

set: LargeAirway1 (p=0.3124)
EXOSC3 | 233495_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7844)
EXOSC3 | 233495_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4013)
EXOSC3 | 233495_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.762)
EXOSC3 | 233495_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7284)
EXOSC3 | 233495_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EXOSC3 | 233495_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0517
p=0.4836
p=0.7058
p=0.6008
p=0.2052
p=0.7934
p=0.8689
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2